VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

Faculty of Computer Science and Engineering

Quiz Review: Performance

Advanced System Architectures

CO5059

Instructor: Tran Ngoc Thinh

Notes by: Phan Nhat Minh (minh.phannhat95@hcmut.edu.vn)

Semester: 261

Lecture 1 – CPU time, CPI, speedup, Amdahl's Law, MIPS/MFLOPS, chip cost and summarizing performance. A cheat sheet, every calculation exercise from the slides with full solutions, the in-class exam question solved, new exam-style questions, and extra practice problems – with solutions at the end.

Last updated: October 4, 2026

Contents

1 Cheat Sheet **1**
- 1.1 Performance and speedup . . . 1
- 1.2 The CPU equation . . . 1
- 1.3 Amdahl's Law . . . 1
- 1.4 MIPS and MFLOPS . . . 2
- 1.5 Cost of an integrated circuit . . . 2
- 1.6 Summarizing several programs . . . 2

2 Exercises from the Slides (with Solutions) **2**

3 Exam-Style Questions **5**

4 Practice Problems **7**

5 Solutions to the Practice Problems **8**

6 Solutions to the Exam-Style Questions **10**

1 Cheat Sheet

1.1 Performance and speedup

Quantity	Formula
Performance	$\text{Perf} = \dfrac{1}{\text{ExTime}}$
Speedup (A over B)	$\dfrac{\text{Perf}_A}{\text{Perf}_B} = \dfrac{\text{ExTime}_B}{\text{ExTime}_A}$ (old / new)
"X is n% faster than Y"	$n = 100 \cdot \dfrac{\text{ExTime}_Y - \text{ExTime}_X}{\text{ExTime}_X}$ (divide by the *faster* time)
CPU time	User CPU + System CPU; $\%\text{CPU} = \dfrac{u+s}{\text{elapsed}}$
Throughput	tasks per unit time (e.g. passengers × mph)

1.2 The CPU equation

Theorem 1.1 (CPU time).

$$T = I \times \text{CPI} \times C = \frac{I \times \text{CPI}}{f}, \qquad C = \frac{1}{f}$$

$$\text{CPU cycles} = \sum_i \text{CPI}_i\, C_i, \qquad \text{CPI} = \frac{\text{CPU cycles}}{I} = \sum_i \text{CPI}_i\, F_i, \qquad F_i = \frac{C_i}{I}$$

$$\text{Fraction of time in class } i = \frac{\text{CPI}_i\, F_i}{\text{CPI}}$$

Changes …	I	CPI	C
Program / algorithm	X	X	
Compiler	X	X	
ISA	X	X	
Organization		X	X
Technology			X

Note. Unit conversions: 200 MHz ⇒ $C = 5$ ns; 1 GHz ⇒ 1 ns; 2.5 GHz ⇒ 0.4 ns. Always write $T = I \cdot \text{CPI}/f$ with f in Hz to avoid mistakes.

1.3 Amdahl's Law

Theorem 1.2 (Amdahl's Law). If a fraction F of the *execution time* is sped up by S:

$$\text{ExTime}_{new} = \text{ExTime}_{old}\left[(1-F) + \frac{F}{S}\right], \qquad \text{Speedup} = \frac{1}{(1-F) + F/S} \le \frac{1}{1-F}.$$

Several disjoint enhancements: $\text{Speedup} = \dfrac{1}{(1 - \sum F_i) + \sum F_i/S_i}$.

Proposition 1.3 (Required enhancement for a target speedup T).

$$\frac{1}{T} = (1-F) + \frac{F}{S} \quad \Longrightarrow \quad S = \frac{F}{\frac{1}{T} - (1-F)}.$$

If $\frac{1}{T} \leq 1 - F$ (i.e. $T \geq \frac{1}{1-F}$), the target is **impossible**.

Remark. **Classic trap:** F is a fraction of *time*, not of instruction count. If you are given an instruction mix, convert first: $F = \text{CPI}_i F_i / \text{CPI}$. Alternatively, skip Amdahl and compare old vs. new CPI directly with the CPU equation – both give the same answer.

1.4 MIPS and MFLOPS

$$\text{MIPS} = \frac{I}{T \times 10^6} = \frac{f}{\text{CPI} \times 10^6} \qquad \text{MFLOPS} = \frac{\#\text{FP ops}}{T \times 10^6}$$

MIPS only compares fairly with the **same program, same ISA, same compiler**. It can rank machines *opposite* to execution time.

1.5 Cost of an integrated circuit

$$\text{Dies per wafer} = \frac{\pi(d/2)^2}{A} - \frac{\pi d}{\sqrt{2A}} \qquad (d = \text{wafer diameter},\ A = \text{die area})$$

$$\text{Die yield} = \text{Wafer yield} \times \left(1 + \frac{D \cdot A}{\alpha}\right)^{-\alpha}, \qquad \alpha \approx 4$$

$$\text{Die cost} = \frac{\text{Wafer cost}}{\text{Dies per wafer} \times \text{Die yield}}$$

$$\text{IC cost} = \frac{\text{Die} + \text{Die test} + \text{Packaging} + \text{Final test}}{\text{Final test yield}}$$

Learning curve: every doubling of volume cuts cost by $\approx 10\%$.

1.6 Summarizing several programs

Arithmetic mean	$\frac{1}{n}\sum T_i$	tracks total time
Weighted mean	$\sum W_i T_i$	result depends on the weights chosen
Geometric mean	$\left(\prod \frac{T_i}{T_{ref,i}}\right)^{1/n}$	same ranking for any reference machine
Harmonic mean	$\frac{n}{\sum 1/R_i}$	for averaging *rates*

2 Exercises from the Slides (with Solutions)

Exercise 2.1 (Unix `time`). The command prints `90.7u 12.9s 2:39 65%`. Interpret it.

Solution. User CPU = 90.7 s, system CPU = 12.9 s, elapsed = 2:39 = 159 s. CPU time = 103.6 s, and 103.6/159 = 65% of the elapsed time. The other 35% is I/O or time spent running other programs. □

Exercise 2.2 (Two notions of performance). Boeing 747: 6.5 h DC–Paris, 610 mph, 470 passengers. Concorde: 3 h, 1350 mph, 132 passengers. Compare latency and throughput.

Solution. Latency: Concorde is $6.5/3 = 1350/610 = 2.2\times$ faster. Throughput (passenger-miles per hour): Boeing $470 \times 610 = 286{,}700$, Concorde $132 \times 1350 = 178{,}200$. Boeing is $286{,}700/178{,}200 = 1.6\times$ higher throughput. Which is "faster" depends on the metric. □

Exercise 2.3 (Price-performance: train vs. plane). Move 2400 people. Train: 4 h trip, 2400 seats, 300 Rs/person. Plane: 45 min trip, 300 seats, 3000 Rs/person. Compare time per person and cost-performance.

Solution. Plane needs $2400/300 = 8$ trips $= 6$ h; train needs 4 h. Time per person: train 4 h/2400 = 6 s, plane 6 h/2400 = 9 s, so the train has 50% better throughput even though each plane trip is $5\times$ faster. Cost $\times$ time: train $300 \times 6 = 1800$, plane $3000 \times 9 = 27{,}000$ Rs·s/person: the train is $15\times$ better. □

Exercise 2.4 (Speedup). A program takes 1 s on machine A and 10 s on machine B. Speedup of A over B?

Solution. Speedup $= \text{ExTime}_B/\text{ExTime}_A = 10/1 = 10$: A is 10 times faster. □

Exercise 2.5 (CPU time). $I = 10{,}000{,}000$ instructions, CPI $= 2.5$, clock 200 MHz. CPU time?

Solution. $C = 1/200$ MHz $= 5 \times 10^{-9}$ s, so $T = 10^7 \times 2.5 \times 5 \times 10^{-9} = 0.125$ s. □

Exercise 2.6 (New compiler and faster CPU). Same program. A new compiler gives $I' = 9{,}500{,}000$ and $\text{CPI}' = 3.0$; the new CPU runs at 300 MHz. Speedup?

Solution. $T' = 9.5 \times 10^6 \times 3.0/(300 \times 10^6) = 0.095$ s. Speedup $= 0.125/0.095 = 1.32$, i.e. 32% faster. □

Exercise 2.7 (Instruction classes). $\text{CPI}_A = 1$, $\text{CPI}_B = 2$, $\text{CPI}_C = 3$. Sequence 1 has counts $(2, 1, 2)$, sequence 2 has $(4, 1, 1)$. Find each CPI. Which is faster?

Solution. Seq. 1: cycles $= 2 + 2 + 6 = 10$, $I = 5$, CPI $= 2.0$. Seq. 2: cycles $= 4 + 2 + 3 = 9$, $I = 6$, CPI $= 1.5$. Sequence 2 is faster (9 vs. 10 cycles) even though it executes *more* instructions. □

Exercise 2.8 (RISC instruction mix). ALU 50% (CPI 1), Load 20% (CPI 5), Store 10% (CPI 3), Branch 20% (CPI 2). Find the CPI and the fraction of time spent in each class.

Solution. CPI $= 0.5 + 1.0 + 0.3 + 0.4 = 2.2$. Time fractions: ALU $0.5/2.2 = 23\%$, Load $1.0/2.2 = 45\%$, Store $0.3/2.2 = 14\%$, Branch $0.4/2.2 = 18\%$. □

Exercise 2.9 ($n\%$ faster). Y takes 15 s, X takes 10 s. How much faster is X?

Solution. $n = 100\,(15 - 10)/10 = 50\%$. □

Exercise 2.10 (Amdahl: floating point). FP instructions become $2\times$ faster, but only 10% of the time is FP. Overall speedup?

Solution. $\text{ExTime}_{new} = (0.9 + 0.1/2)\,\text{ExTime}_{old} = 0.95\,\text{ExTime}_{old}$, so Speedup $= 1/0.95 = 1.053$. □

Exercise 2.11 (Amdahl: load CPI). In the RISC mix above, load CPI improves from 5 to 2. Speedup?

Solution. Loads take $F = 45\%$ of the time (not 20%), $S = 5/2 = 2.5$: Speedup $= 1/(0.55 + 0.45/2.5) = 1/0.73 = 1.37$. Check with CPI: new CPI $= 0.5 + 0.2 \times 2 + 0.3 + 0.4 = 1.6$, and $2.2/1.6 = 1.37$. □

Exercise 2.12 (How fast must multiply be?). A program runs in 100 s, of which multiply takes 80 s. How much faster must multiply be to make the program (a) $4\times$ faster, (b) $5\times$ faster?

Solution. (a) Target 25 s: $25 = 20 + 80/n \Rightarrow n = 16$. (b) Target 20 s: $20 = 20 + 80/n \Rightarrow 80/n = 0$: impossible. The unaffected 20 s already equals the target ($1/(1-F) = 1/0.2 = 5$ is the limit and cannot be reached). □

Exercise 2.13 (Multiple enhancements). $S_1 = 10, F_1 = 20\%$; $S_2 = 15, F_2 = 15\%$; $S_3 = 30, F_3 = 10\%$ (disjoint parts). Overall speedup?

Solution. Speedup $= 1/\big(0.55 + 0.02 + 0.01 + 0.0033\big) = 1/0.5833 = 1.71$. □

Exercise 2.14 (MIPS can mislead). Clock 100 MHz; CPI of classes A, B, C $= 1, 2, 3$. Compiler 1 produces $(5, 1, 1)$ million instructions; compiler 2 produces $(10, 1, 1)$ million. Com-

pare MIPS and CPU time.

Solution. Compiler 1: CPI $= 10/7 = 1.43$, MIPS $= 100/1.43 = 70$, time $= 7 \times 10^6 \times 1.43/10^8 = 0.10$ s. Compiler 2: CPI $= 15/12 = 1.25$, MIPS $= 80$, time $= 12 \times 10^6 \times 1.25/10^8 = 0.15$ s. Compiler 2 has the higher MIPS but is *slower*. □

Exercise 2.15 (Dies per wafer). Die 0.7 cm on a side, wafer diameter 30 cm. How many dies?

Solution. $A = 0.49$ cm^2: $\pi(15)^2/0.49 - \pi \cdot 30/\sqrt{0.98} = 1442.6 - 95.2 \approx 1347$ dies. □

Exercise 2.16 (Die yield). Same die, defect density $0.6/\text{cm}^2$, $\alpha = 4$, wafer yield 100%.

Solution. $(1 + 0.6 \times 0.49/4)^{-4} = 1.0735^{-4} = 0.75$. □

Exercise 2.17 (Weighted means). P1/P2 take 1/1000 s on A, 10/100 s on B, 20/20 s on C. Rank the machines under weights $(0.5, 0.5)$, $(0.909, 0.091)$, $(0.999, 0.001)$.

Solution.

Weights	A	B	C
.5/.5	500.5	55.0	**20**
.909/.091	91.91	**18.19**	20
.999/.001	**2.0**	10.09	20

Each weighting picks a different winner (C, then B, then A). (The slide prints 91.82 and 18.8 for the middle row; the exact values are 91.91 and 18.19 – the ranking is the same.) □

3 Exam-Style Questions

The first question was done in class; its solution follows it directly. The others are new questions in the same style – their solutions are in the last section.

Exercise 3.1 (In class – two implementations of one ISA). An instruction set has four classes A, B, C, D. Implementation M1 runs at 2.4 GHz and M2 at 3.0 GHz. Cycles per instruction and class frequencies in a typical program:

Class	M1 cycles/instr.	M2 cycles/instr.	Frequency
A	2	3	20%
B	8	4	50%
C	1	10	10%
D	4	2	20%

(a) Calculate the average CPI for M1.

(b) Calculate the MIPS rating for M1.

(c) Calculate the average CPI for M2.

(d) Calculate the MIPS rating for M2.

(e) A program executes 23.7×10^9 instructions (on either implementation). How long does it take on M1, to the nearest millisecond?

(f) Repeat (e) for M2.

Solution.

(a) $\text{CPI}_{M1} = 0.2(2) + 0.5(8) + 0.1(1) + 0.2(4) = 0.4 + 4.0 + 0.1 + 0.8 = 5.3.$

(b) $\text{MIPS}_{M1} = \dfrac{f}{\text{CPI} \times 10^6} = \dfrac{2.4 \times 10^9}{5.3 \times 10^6} = 452.8.$

(c) $\text{CPI}_{M2} = 0.2(3) + 0.5(4) + 0.1(10) + 0.2(2) = 0.6 + 2.0 + 1.0 + 0.4 = 4.0.$

(d) $\text{MIPS}_{M2} = \dfrac{3.0 \times 10^9}{4.0 \times 10^6} = 750.$

(e) $T_{M1} = \dfrac{I \times \text{CPI}}{f} = \dfrac{23.7 \times 10^9 \times 5.3}{2.4 \times 10^9} = 52.3375 \text{ s} \approx 52{,}338 \text{ ms}.$

(f) $T_{M2} = \dfrac{23.7 \times 10^9 \times 4.0}{3.0 \times 10^9} = 31.6 \text{ s} = 31{,}600 \text{ ms}.$

M2 is $52.3375/31.6 = 1.66\times$ faster. Here the MIPS ratio ($750/452.8 = 1.66$) agrees with the time ratio only because both machines run the *same* instruction count (same ISA, same program). □

Exercise 3.2 (Two processors, one program). An ISA has three classes. Processor P1 runs at 2.0 GHz and P2 at 2.5 GHz:

Class	P1 CPI	P2 CPI	Frequency
A	1	2	50%
B	3	2	30%
C	5	6	20%

(a) Average CPI of P1.

(b) MIPS rating of P1.

(c) Average CPI of P2.

(d) MIPS rating of P2.

(e) A program executes 15×10^9 instructions. Find its execution time on each processor. Which is faster, and by what percentage?

(f) P1's designers reduce the CPI of class C from 5 to 3. Find P1's new execution time and check the speedup with Amdahl's Law. Which processor is faster now?

Exercise 3.3 (Two compilers). A 3 GHz processor has classes A, B, C with CPI 1, 2 and 4.

Two compilers produce these instruction counts (in billions) for the same program:

Compiler	A	B	C
X	6	2	2
Y	12	2	1.5

(a) Total instruction count and average CPI for each compiler.

(b) MIPS rating for each compiler.

(c) CPU time for each compiler.

(d) Which compiler would a buyer choose by MIPS, and which is really faster? By what percentage?

(e) A hardware change halves the CPI of class C ($4 \to 2$). For compiler X's code, find the new CPU time.

(f) Verify (e) with Amdahl's Law.

Exercise 3.4 (Cost of a die). A 30 cm wafer costs \$6000. The die area is 1.2 cm^2, the defect density 0.5/cm^2, $\alpha = 4$, and the wafer yield is 100%.

(a) How many dies fit on the wafer?

(b) What is the die yield?

(c) How many good dies does a wafer produce?

(d) What is the cost per good die?

(e) A new process shrinks the die to 0.8 cm^2 but the wafer costs \$7000. Repeat (a)–(d).

(f) Explain why the cost per die falls much more than the area.

4 Practice Problems

Try these before looking at the solutions in the next section.

Exercise 4.1. A program executes 2×10^9 instructions with CPI = 1.5 on a 2.5 GHz processor. What is its CPU time?

Exercise 4.2. The same program runs on processor B with CPI = 1.2 at 2.2 GHz. Which processor is faster, and by how much?

Exercise 4.3. A program runs in 10 s on a 2 GHz computer A. You are designing computer B to run it in 6 s. The faster clock forces B to need 1.2× as many clock cycles as A. What clock rate must B have?

Exercise 4.4. Classes A, B, C have CPI 1, 2, 3; clock 1 GHz. Compiler X emits (4, 2, 1) million instructions of each class; compiler Y emits (2, 2, 2) million. For each: instruction

count, CPI, CPU time, MIPS. Which is faster?

Exercise 4.5. ALU 45% (CPI 1), Load 25% (CPI 4), Store 10% (CPI 3), Branch 20% (CPI 2). Find the CPI and the fraction of time spent in each class.

Exercise 4.6. For the mix in Problem 4.5, you can either (a) reduce load CPI from 4 to 2, or (b) reduce branch CPI from 2 to 1. Which gives the larger speedup?

Exercise 4.7. 40% of a program's time is vectorizable and a vector unit runs that part 10× faster. What is the overall speedup? What is the best possible speedup from vectorization alone?

Exercise 4.8. FP code is 70% of execution time. How much faster must the FP unit be for an overall speedup of 2? Of 3.5?

Exercise 4.9. Enhancement E1 speeds up 20% of the time by 10×; E2 speeds up 50% of the time by 1.6× (different parts). Which one alone is better? What do you get with both?

Exercise 4.10. (a) X takes 8 s and Y takes 12 s. How much faster is X? (b) X is 25% faster than Y, and Y takes 10 s. How long does X take?

Exercise 4.11. A new compiler reduces instruction count by 20% but raises CPI by 10%. Speedup?

Exercise 4.12. A 500 MHz machine runs a mix of 60% (CPI 1), 30% (CPI 2), 10% (CPI 4). Find CPI, MIPS rating, and the time for 10^9 instructions.

Exercise 4.13. `time` prints `45.3u 4.7s 1:40 50%`. Check the 50% and find what fraction of CPU time is user time.

Exercise 4.14. Wafer diameter 20 cm, wafer cost \$1000, defect density 0.8/cm^2, $\alpha = 4$, wafer yield 100%. Find dies per wafer, die yield and die cost for (a) a 1 cm^2 die and (b) a 0.5 cm^2 die.

Exercise 4.15. Times (s) for P1, P2: machine A $(2, 20)$, machine B $(4, 10)$, reference R $(4, 20)$. Compare A and B by total time and by geometric mean of normalized times. Repeat the geometric mean using A as the reference.

5 Solutions to the Practice Problems

Solution (4.1). $T = 2 \times 10^9 \times 1.5/(2.5 \times 10^9) = 1.2$ s. □

Solution (4.2). $T_B = 2 \times 10^9 \times 1.2/(2.2 \times 10^9) = 1.091$ s. B is faster: $1.2/1.091 = 1.10$, i.e. 10% faster. □

Solution (4.3). Cycles on A $= 10 \times 2 \times 10^9 = 2 \times 10^{10}$. B needs 2.4×10^{10} cycles in 6 s: $f_B = 2.4 \times 10^{10}/6 = 4$ GHz. □

Solution (4.4). X: $I = 7$M, cycles $= 4 + 4 + 3 = 11$M, CPI $= 1.57$, $T = 11$ ms, MIPS $= 1000/1.57 = 636$. Y: $I = 6$M, cycles $= 2 + 4 + 6 = 12$M, CPI $= 2.0$, $T = 12$ ms, MIPS $= 500$. X is faster ($12/11 = 1.09\times$) even though it executes more instructions. □

Solution (4.5). CPI $= 0.45 + 1.00 + 0.30 + 0.40 = 2.15$. Time fractions: ALU 20.9%, Load 46.5%, Store 14.0%, Branch 18.6%. □

Solution (4.6). (a) New CPI $= 2.15 - 0.25 \times 2 = 1.65$, speedup $= 2.15/1.65 = 1.30$. (b) New CPI $= 2.15 - 0.20 \times 1 = 1.95$, speedup $= 1.10$. Option (a) wins: loads are the common case in *time* (46.5%). Amdahl check for (a): $1/(0.535 + 0.465/2) = 1.30$. □

Solution (4.7). $1/(0.6 + 0.4/10) = 1/0.64 = 1.56$. Even an infinitely fast vector unit gives at most $1/0.6 = 1.67$. □

Solution (4.8). For 2: $0.5 = 0.3 + 0.7/S \Rightarrow S = 3.5$. For 3.5: $1/3.5 = 0.286 < 0.3$, impossible; the maximum is $1/0.3 = 3.33$. □

Solution (4.9). E1: $1/(0.8 + 0.02) = 1.22$. E2: $1/(0.5 + 0.3125) = 1.23$. E2 is (slightly) better. Both: $1/(0.3 + 0.02 + 0.3125) = 1/0.6325 = 1.58$. □

Solution (4.10). (a) $n = 100 \times (12 - 8)/8 = 50\%$. (b) $\text{ExTime}_Y/\text{ExTime}_X = 1.25 \Rightarrow \text{ExTime}_X = 10/1.25 = 8$ s (not 7.5 s). □

Solution (4.11). $T'/T = 0.8 \times 1.1 = 0.88$, so speedup $= 1/0.88 = 1.14$. □

Solution (4.12). CPI $= 0.6 + 0.6 + 0.4 = 1.6$; MIPS $= 500/1.6 = 312.5$; $T = 10^9 \times 1.6/(5 \times 10^8) = 3.2$ s ($= 10^9/312.5 \times 10^6$). □

Solution (4.13). Elapsed $= 100$ s; $(45.3 + 4.7)/100 = 50\%$. User share of CPU time $= 45.3/50 = 90.6\%$. □

Solution (4.14). (a) Dies $= \pi \cdot 100/1 - \pi \cdot 20/\sqrt{2} = 314.2 - 44.4 = 269$. Yield $= (1 + 0.8/4)^{-4} = 1.2^{-4} = 0.482$. Die cost $= 1000/(269 \times 0.482) = \$7.71$. (b) Dies $= 628.3 - 62.8 = 565$. Yield $= 1.1^{-4} = 0.683$. Die cost $= 1000/(565 \times 0.683) = \$2.59$. Halving the area cuts the die cost by about $3\times$: more dies fit *and* each is less likely to contain a defect. □

Solution (4.15). Total time: A = 22 s, B = 14 s, so B is better for this workload. Normalized to R: A $= (0.5, 1.0)$, B $= (1.0, 0.5)$; both geometric means are $\sqrt{0.5} = 0.707$, a tie. Normalized to A: A $= (1, 1) \to 1$, B $= (2, 0.5) \to 1$, again a tie. The geometric mean is consistent across references, but it does not track total execution time. □

6 Solutions to the Exam-Style Questions

Solution (3.2).

(a) $\text{CPI}_{P1} = 0.5(1) + 0.3(3) + 0.2(5) = 0.5 + 0.9 + 1.0 = 2.4$.

(b) $\text{MIPS}_{P1} = 2000/2.4 = 833.3$.

(c) $\text{CPI}_{P2} = 0.5(2) + 0.3(2) + 0.2(6) = 1.0 + 0.6 + 1.2 = 2.8$.

(d) $\text{MIPS}_{P2} = 2500/2.8 = 892.9$.

(e) $T_{P1} = 15 \times 10^9 \times 2.4/(2 \times 10^9) = 18$ s; $T_{P2} = 15 \times 10^9 \times 2.8/(2.5 \times 10^9) = 16.8$ s. P2 is faster: $18/16.8 = 1.071$, i.e. 7.1% faster.

(f) New $\text{CPI}_{P1} = 0.5 + 0.9 + 0.2(3) = 2.0$, $T = 15$ s, speedup $18/15 = 1.2$. Amdahl: class C took $1.0/2.4 = 41.7\%$ of the time, $S = 5/3$: $1/(0.583 + 0.417 \times 0.6) = 1/0.833 = 1.2$. P1 is now faster than P2 by $16.8/15 = 1.12$ (12%).

□

Solution (3.3).

(a) X: $I = 10 \times 10^9$, cycles $= 6 + 4 + 8 = 18 \times 10^9$, CPI = 1.8. Y: $I = 15.5 \times 10^9$, cycles $= 12 + 4 + 6 = 22 \times 10^9$, CPI $= 22/15.5 = 1.42$.

(b) X: $3000/1.8 = 1667$ MIPS. Y: $3000/1.42 = 2114$ MIPS.

(c) X: $18 \times 10^9/(3 \times 10^9) = 6$ s. Y: $22 \times 10^9/(3 \times 10^9) = 7.33$ s.

(d) MIPS favours Y, but X is faster: $7.33/6 = 1.22$, i.e. 22% faster. Y executes many more (cheap) instructions, which raises MIPS but also the total time.

(e) Cycles $= 6 + 4 + 2 \times 2 = 14 \times 10^9$, $T = 4.67$ s.

(f) Class C took $8/18 = 44.4\%$ of X's time, $S = 2$: $1/(0.556 + 0.222) = 1.286 = 6/4.67$. ✓

□

Solution (3.4).

(a) $\dfrac{\pi(15)^2}{1.2} - \dfrac{\pi \cdot 30}{\sqrt{2.4}} = 589.0 - 60.8 = 528$ dies.

(b) $(1 + 0.5 \times 1.2/4)^{-4} = 1.15^{-4} = 0.572$.

(c) $528 \times 0.572 \approx 302$ good dies.

(d) $6000/301.9 \approx \$19.9$.

(e) Dies $= 883.6 - 74.5 = 809$; yield $= (1 + 0.1)^{-4} = 0.683$; good dies ≈ 553; cost $= 7000/552.6 \approx \$12.7$.

(f) A smaller die gives more dies per wafer (fewer wasted at the edge, too) *and* a higher yield, because each die is less likely to contain a defect. The two effects multiply, so cost falls faster than area (here a 33% smaller die is 36% cheaper even though the wafer costs 17% more).

□